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FIFO Page Replacement Algorithm

Write a C program to implement the FIFO Page Replacement Algorithm. The program should take the number of frames and a reference string (sequence of page requests) as input. For each page in the reference string, determine if it results in a 'Page Hit' or a 'Page Fault'. Display the state of the frames after each request and calculate the total number of page faults and hits.

Input Format

1. The first line of input contains an integer representing the number of pages in the reference string.
2. The second line contains the page reference string as a sequence of integers.
3. The third line contains an integer representing the number of available frames.

Output Format

The total number of Page Faults and Page Hits are displayed

Sample Input 1

Select Language: C (GCC 9.2.0)

```
1 #include <stdio.h>
2
3 int main() {
4     int n, frames_count, i, j;
5     int page_faults = 0, page_hits = 0, pointer = 0;
6
7     printf("Enter number of pages in reference string: \n");
8     if (scanf("%d", &n) != 1) return 0;
9
10    int reference_string[n];
11    printf("Enter the reference string: \n");
12    for (i = 0; i < n; i++) {
13        scanf("%d", &reference_string[i]);
14    }
15
16    printf("Enter number of frames: \n");
17    if (scanf("%d", &frames_count) != 1) return 0;
18
19    int frames[frames_count];
20
21    // Initialize frames with -1 to represent empty slots
22    for (i = 0; i < frames_count; i++) {
23        frames[i] = -1;
24    }
```

Test Cases 2

Run Sample Cases

Run All Cases

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Sample Input 1

Select Language : C (GCC 9.2.0)

```

26  for (i = 0; i < n; i++) {
27      int found = 0;
28
29      // Check for Page Hit
30      for (j = 0; j < frames_count; j++) {
31          if (frames[j] == reference_string[i]) {
32              found = 1;
33              page_hits++;
34              break;
35          }
36      }
37
38      // Handle Page Fault
39      if (!found) {
40          frames[pointer] = reference_string[i];
41          pointer = (pointer + 1) % frames_count; // FIFO Logic: Replace oldest
42          page_faults++;
43      }
44  }
45
46  printf("\nTotal Page Faults: %d", page_faults);
47  printf("\nTotal Page Hits: %d\n", page_hits);
48
49  return 0;
50  }

```

Test Cases 2

Run Sample Cases

Run All Cases